

# **Introduction to quantum computing**

August 13, 2026

Serena Chan

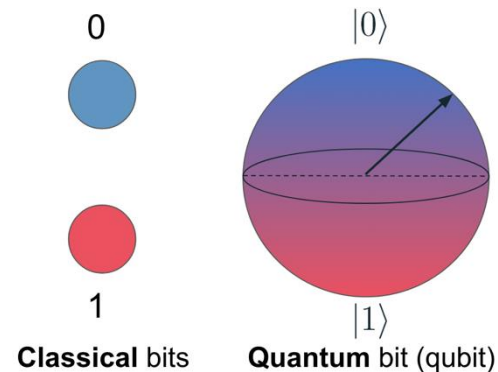
# What is quantum computing?

- Derived from quantum physics: the physical properties of nature at the scale of atoms and subatomic particles\ul>  - Energy is quantized; not continuous
  - Objects can behave as both waves and particles, exist in multiple states at once, and are governed by probabilities rather than certainties
- Classical computing:
  - Deterministic
  - Uses binary states (bits)
- Quantum computing:
  - Probabilistic
  - Uses quantum states (qubits)

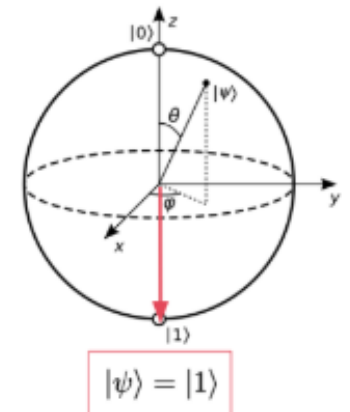
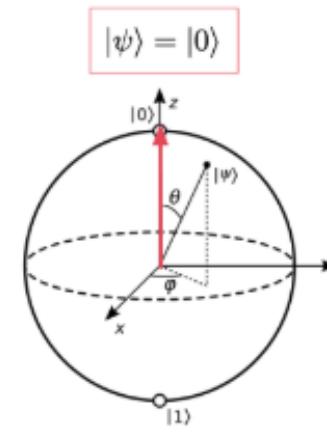
# Quantum bits (qubits)

- The fundamental unit of information of a quantum computer is the qubit, mathematically represented by state vectors.
- A qubit's state is represented by a vector in a complex vector space, written as a linear combination of the basis states  $|0\rangle$  and  $|1\rangle$ 
  - This is called superposition:  $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$  with  $|\alpha|^2 + |\beta|^2 = 1$
- Wherein a ket  $|\cdot\rangle$  is a column vector and its associated bra  $\langle\cdot|$  is a row vector
  - Ex.  $|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ ,  $|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$        $\langle 0| = [1 \ 0]$ ,  $\langle 1| = [0 \ 1]$ .

- Visually, we can see it like this:

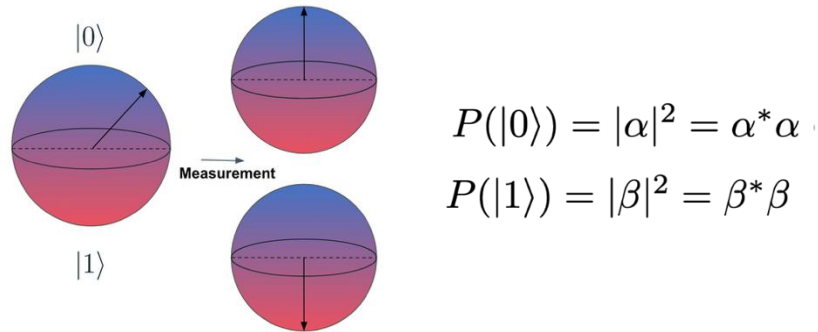


where

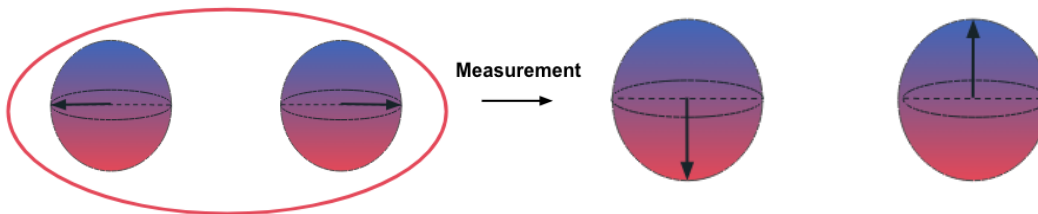


# Qubit properties/operations

- Probabilistic measurements:
  - Although qubits are linear combination of states, when measured it will only be in the classical states, 0 or 1



- Entanglements:
  - Entangled qubits are correlated, wherein measuring one changes the measurement of others



**Demo #1**

# Multiple qubits

- For a system with multiple qubits, the register is created by performing a tensor product of all states on the computational basis

• Ex.

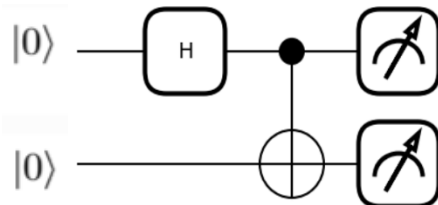


$$|0\rangle \otimes |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ 0 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ \begin{bmatrix} 0 \\ 0 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

- Where a system with n qubits is  $|q_0\rangle \otimes |q_1\rangle \otimes \cdots \otimes |q_{n-1}\rangle$ .
- As qubits are added, the state space grows, where dimension =  $2^n$
- However, some states cannot be written as the product of individual states, aka entanglement.
- Ex.  $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \neq |\text{qubit 1}\rangle \otimes |\text{qubit 2}\rangle$

# Algorithms/circuits

- A quantum algorithm manipulates the state of qubits to solve some problem.
  1. Prepare qubits in a superposition
  2. Apply operations
    - Operations that create entanglement and interference
    - Gates chosen such that amplitudes of wrong answers cancel, and amplitudes of right answers reinforce
  3. Measure qubits and extract solutions
- Quantum circuits are implementations of the algorithms, involving sequences of gates applied to the qubits across a timeline such that a machine can run it.



**Demo #2**



# Applications

- What can the qubit encode?
  1. A spin-orbital (occupied/empty)
    - Simulation of molecular electronic structure
  2. A binary choice in a combinatorial problem
    - Optimization wherein answer is a bitstring
  3. A coordinate in feature space
    - Encoding a high-dimensional vector into quantum space for ML
- Biological and chemical interactions are inherently quantum mechanical at the subatomic level - a quantum circuit natively replicates this rather than trying to artificially approximate them.

# Some recent examples

## 1. [Computation of electronic energies](#)

- Since classical methods only approximate electron correlation, quantum entanglement can allow for better representation of hydrogen bonding and base stacking
- Not bad but difficult to scale and noisy

## 2. [RNA folding prediction](#)

- Each qubit encodes whether or not a specific sequence of paired bases is present in the final folded structure
- Usually only possible with small and does not outperform classical methods

## 3. [Generative models](#)

- Amplitude or angle encoding converts vectors into qubits, then a parameterized quantum circuit will define a probability distribution for a classical optimizer to backprop
- Encoding is super heavy in memory/compute and does not outperform classical methods

# Limitations

- Current hardware is NISQ (Noisy Intermediate Scale Quantum Computers)
  - Qubits are noisy
  - Systems are restricted by a certain # of physical qubits
  - Thus, circuits must be kept as short as possible and can only perform a limited number of unit operations (or gates) before the information degrades.
- In biology:
  - Systems are complex.
  - To achieve true accuracy, we need billions/trillions of shots; which is not possible at scale.



**Content from:**

- CIQ101 and CIQ102 by Calcul Quebec
- <https://pennyLane.ai/codebook/introduction-to-quantum-computing/all-about-qubits>
- <https://docs.pennyLane.ai/en/stable/code/api/pennyLane.RX.html>
- <https://journals.aps.org/prxlife/abstract/10.1103/h49j-bsc6>